

Cleanify: The Smart Batteryless Waste Management System

Abstract—Efficient municipal solid waste management is critical for sustainable urban living, yet existing collection methods often rely on rigid schedules that lead to inefficient routes, unnecessary fuel consumption, and overflowing bins. Prior IoT-based solutions have introduced sensor-equipped bins, but most depend on batteries, creating maintenance burdens and environmental concerns. To overcome these challenges, we present Cleanify, a comprehensive batteryless smart waste management platform that combines energy-harvesting IoT sensors with an advanced software ecosystem. On the hardware side, Cleanify eliminates the need for batteries by leveraging ultra-low-power sensing and wireless communication. On the software side, our system integrates real-time monitoring and cloud-based data management. The result is a scalable, cost-effective, and sustainable platform that reduces maintenance requirements, optimizes fleet operations, and lowers carbon emissions. Field trials demonstrate the feasibility of batteryless IoT for waste monitoring, while the new software stack enables seamless deployment and management in real-world urban environments.

Index Terms—Energy harvesting, IoT, waste management, smart bins, kinetic energy, LoRaWAN

I. INTRODUCTION

Municipalities in low- and middle-income countries (LMICs) spend up to 85% of their solid waste management budgets on collection alone, a cost driven by fixed-route truck fleets that service bins indiscriminately [1], [2]. This brute-force approach results in a costly paradox: trucks waste fuel and labor on half-empty containers while other bins overflow, creating public health crises by fostering disease vectors for infections like diphtheria and dysentery [3], [4]. While IoT-enabled “smart bins” with fill-level sensors promise data-driven efficiency gains [5]–[7], their adoption in resource-constrained regions is crippled by a critical flaw: the maintenance burden of battery-powered sensors. At city scale, periodically recharging or replacing batteries in thousands of bins is logistically and economically infeasible for already-strained authorities.

This reality demands a paradigm shift from battery-powered to entirely self-sustaining sensors [8]. While solar harvesting has been explored [9], it is an unreliable solution for urban bins frequently shaded by buildings or installed indoors. We argue that a far more ubiquitous and reliable power source is already available: *the kinetic energy from the routine opening and closing of a bin’s lid*. This source is intrinsically tied to the bin’s usage pattern, making it a perfect candidate for energy harvesting.

This paper introduces Cleanify, an end-to-end smart waste management system designed around a novel batteryless and retrofit sensor node (see Figure 1). The system targets standardized lid or hinge based municipal-scale waste bins which experience frequent lid actuation throughout the day. The node harvests kinetic energy from these lid movements

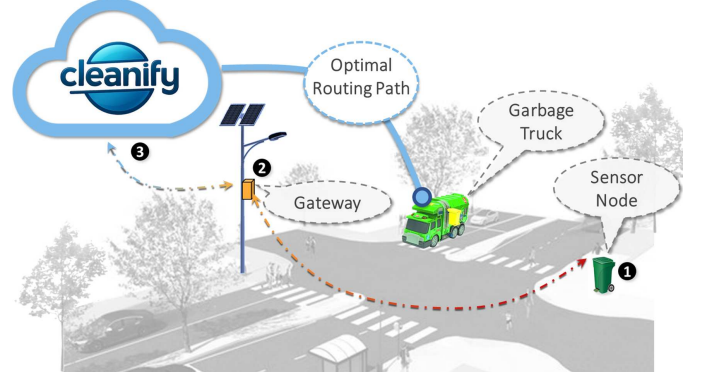


Fig. 1. Cleanify smart waste collection ecosystem.

to power an ultrasonic sensor and a LoRaWAN transceiver, enabling maintenance-free operation [10], [11]. It is designed as a “plug-and-play” module that can be attached to existing municipal bins without requiring costly replacements or structural modifications. While our system includes a lightweight cloud backend for data aggregation and routing, the core contribution of this work is the design, implementation, and field validation of the batteryless sensing hardware.

Our path to a viable batteryless sensor was one of iterative, deployment-driven engineering. Our initial prototypes included an external gear based mechanism and a lid-integrated pendulum, both of which failed under real-world conditions due to mechanical friction, low energy conversion, and poor durability. These failures informed the design of our final, miniaturized shaft-based harvester, which overcomes these issues with simplified kinematics and reduced wear. This paper documents the lessons learned from this evolutionary process.

Contributions. We make the following contributions:

- We design, implement, and evaluate a portable, batteryless smart bin sensor that harvests kinetic energy from lid motion to power its sensing and communication, built entirely from low-cost, commodity components.
- We document the evolution of three hardware prototypes, demonstrating a deployment-driven design process where lessons from real-world failures informed a final, robust, and maintenance-free architecture suitable for LMICs.
- We demonstrate the feasibility of our approach through a week-long pilot deployment on a university campus using standard municipal-scale dustbins, validating that Cleanify can operate reliably on harvested energy alone and support more efficient waste collection logistics.

II. RELATED WORK

Efficient waste management has emerged as a critical challenge for urban sustainability, prompting research efforts

spanning IoT-enabled monitoring, energy harvesting technologies, and optimization algorithms. This section surveys relevant work across smart waste monitoring systems, energy harvesting approaches, batteryless architectures, and commercial deployments.

A. IoT-Enabled Smart Waste Management

Early smart bin systems employed ultrasonic or infrared sensors transmitting data via cellular networks or Wi-Fi [5], [6], [12], [13]. However, these approaches face critical limitations in LMICs: cellular subscriptions incur recurring costs, Wi-Fi requires proximate infrastructure, and both consume substantial power necessitating frequent battery replacements.

Low-Power Wide-Area Networks (LPWANs) have emerged as alternatives [14]. Nordsense achieves 6–7 year battery life through aggressive duty cycling [7], while Sensoneo claims 5-year operation on lithium batteries [15]. Despite improved energy efficiency, these solutions remain battery-dependent, creating maintenance burdens at city scale. Route optimization algorithms have demonstrated 36.8% reductions in collection distances [16], but presuppose reliable sensor data undermined by battery failures.

B. Energy Harvesting for IoT Applications

Solar photovoltaics represent the most mature approach, with Bigbelly’s solar-compacting bins achieving autonomous operation [17]. However, solar harvesting suffers from context-dependent reliability: bins installed indoors or in heavily shaded areas receive insufficient irradiance. Bigbelly’s hybrid solar-battery systems still require 3–4 year battery replacement cycles.

Kinetic energy harvesting has been explored for industrial monitoring [18]. Piezoelectric transducers generate μW – mW power [18], [19], struggling to support radio transmission without extensive buffering. Electromagnetic generators offer higher power density [20] but require substantial mechanical displacement—the condition available in bin lid actuation. Asenov and Tokmakov [9] demonstrated batteryless LoRaWAN sensing using solar harvesting, though solar dependency limits indoor applicability.

C. Gap Analysis

Existing approaches face fundamental trade-offs: battery-powered systems impose maintenance burdens; solar-harvesting requires favorable conditions; integrated bins demand high capital investment. Our work addresses this gap through kinetic energy harvesting with plug-and-play retrofitting and commodity hardware. Unlike solar approaches, lid-actuated harvesting functions reliably indoors. Unlike integrated systems, our retrofit module preserves existing infrastructure.

Cleanify represents the first batteryless smart waste system that: (1) harvests energy exclusively from user interactions, (2) operates without solar dependency, (3) retrofits existing bins without modification, and (4) validates through extensive real-world deployment (2,539 actuations).

III. SYSTEM DESIGN

Cleanify proposes a three-tier IoT system (Figure 1). At the edge, our batteryless energy-harvesting sensor nodes monitor bin fill-levels by harvesting kinetic energy from lid actuation. Data is transmitted via LoRaWAN to gateways, which forward it to a cloud backend. The backend aggregates time-series data and applies basic heuristics (e.g., fill-level thresholds) to support downstream applications such as trend prediction, optimal collection scheduling, and fleet routing optimization [21]–[23]. Figure 2 illustrates the envisioned software pipeline for dynamic route generation based on real-time fill-level data. While the hardware platform validated in this work provides the requisite sensor infrastructure, full implementation and empirical validation of the route optimization layer constitutes ongoing work. Since the architecture is modular, such software enhancements can be integrated without altering the hardware, which remains the focus of this paper.



Fig. 2. Envisioned route optimization pipeline using real-time fill-level data. Full implementation constitutes future work.

A. Deployment-Driven Goals

Our design is guided by the operational and economic realities of waste management in LMICs. These constraints translate into three core design goals:

- [D1] **Zero-Maintenance Operation:** Nodes must be fully autonomous, requiring no battery replacements or other physical servicing post-deployment to be scalable.
- [D2] **Plug-and-Play Retrofitting:** The module must easily attach to diverse, existing municipal waste bin infrastructures, particularly standard bins with hinged or foot-pedal-operated lids, without structural modification, avoiding the high capital cost of replacing legacy assets.
- [D3] **Low-Cost Commodity Hardware:** To ensure replicability and local repairability, all components must be inexpensive and widely available in local markets.

B. Batteryless Sensor Node Design

The core of Cleanify is a sensor node engineered around a single, powerful principle: **the event that necessitates a**

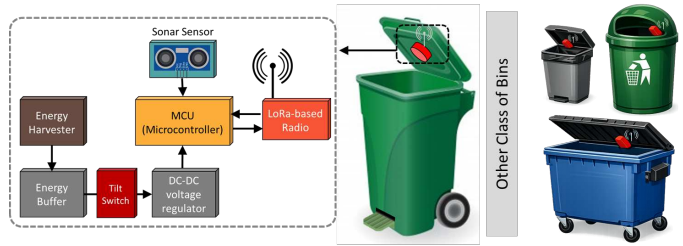
Energy Harvesting, Gating, and Storage: The system’s viability hinges on capturing and storing transient kinetic energy while cleanly separating the *harvesting* and *use* phases. During a lid-opening event, an electromagnetic energy harvester generates a short burst of power. A tilt switch between the harvester and the rest of the circuitry acts as an event-triggered power gate: while the lid is moving, it isolates the electronics so that all harvested energy is directed into the storage element; once the lid returns to its resting position, the switch connects the charged buffer to the downstream regulator and digital logic.

Our design is not specific to a single dustbin geometry. Any receptacle with a routinely actuated component (lid, flap, pedal linkage, or handle) can be instrumented to harvest actuation energy and trigger a sensing/transmission cycle, with the mechanical interface being the primary element that changes across form factors.

The primary design choice in this block is the energy storage medium. We deliberately avoided rechargeable batteries, as their limited charge/discharge cycles and susceptibility to environmental stress make them a long-term maintenance liability. Instead, we opted for a capacitor-based energy buffer. A capacitor offers virtually unlimited charge cycles and high power-delivery rates, making it well suited to our use case: accepting a rapid charge from the harvester and then discharging quickly to power a LoRa radio transmission. This combination of event-triggered gating and capacitor storage is fundamental to achieving true “fit-and-forget” longevity [D1].

Fill-Level Sensing Modality: To determine bin fullness, we evaluated two distinct sensing paradigms: weight-based and range-based.

- **Weight-based sensing**, using load cells at the bin’s base, was rejected early in our design. This approach is fundamentally flawed for waste management, as it conflates weight with volume; a half-full bin of dense materials could trigger a false “full” alert. Furthermore, it would require significant structural modification to existing bins, violating our plug-and-play retrofitting goal [D2].
- **Range-based sensing**, which measures the distance from the lid to the waste surface, directly measures remaining volume. This method is non-intrusive and aligns perfectly with our retrofitting goal. Within this paradigm, we favored **ultrasonic ranging** over optical or infrared time-of-flight sensors. Ultrasonic sensors provide a wider sensing cone, making them more robust to irregular trash topographies, and



are significantly lower in cost, adhering to our commodity-hardware goal [D3].

Computation and Communication: The node’s computational and communication strategy is built for extreme energy scarcity. The system requires a low-power microcontroller capable of managing the sensing and communication cycle on a meager energy budget harvested from a single event. For communication, the clear choice was a **Low-Power Wide-Area Network (LPWAN)** protocol. Unlike power-hungry cellular or short-range Wi-Fi/BLE, LPWANs are designed for transmitting small data packets over many kilometers. We selected **LoRaWAN** for its open standard, resilience to urban interference, and the wide availability of low-cost transceiver modules [24], which is critical for scalability and replicability [D3].

Before committing to a full-scale prototype design, we conducted a series of preliminary micro-evaluations to answer critical questions about the viability of our battery-less waste-monitoring approach. These focused experiments allowed us to validate core assumptions and inform key design parameters early on. In particular, we sought to determine whether everyday bin usage could generate sufficient energy to power a sensor node. Resolving these uncertainties before building an integrated system helped us avoid costly dead-ends. The two central questions we posed were:

Q1: How much energy do we need to harvest to run the system reliably?

Rationale: We first needed to establish a precise energy budget for a complete sensing-and-transmission operation, including powering an ultrasonic sensor, an MCU, and a LoRa radio module. Answering this question was essential to set a clear target for the energy harvesting mechanism..

Q2: How much energy can be harvested from a single bin-lid movement? *Rationale:* The system’s viability hinges on extracting sufficient energy from the mechanical motion of the bin’s lid. We needed to characterize typical user interactions (e.g., the speed and angle of lid openings) to estimate the available kinetic energy and determine if it could meet the budget established in Q1.

We divide our edge node implementation into two major subsystems: the **Power Management System**, and **Form Factor & Isolation**.

1) Power Management System:

a) *Power Management Policy:* Minimizing power draw was a key requirement. The ATmega328P was configured to operate using its internal RC oscillator with all auxiliary peripherals (including the Brown-Out Detector) disabled [25] to reduce passive consumption.

The RA-02 module exhibits high current spikes during transmission but remains in deep sleep otherwise [26].

The HC-SR04 poses the greatest challenge, as its design lacks a low-power mode [27]. While transistors could disable it between measurements, experiments revealed that parasitic power through the trigger and echo pins allowed partial operation even when switched off.

A more complex isolation approach could mitigate this, but it would introduce nonstandard components and contradict our goal of using easily available parts.

b) *Energy Harvesting Compatibility:* To ensure versatility, the node was designed to accept power from multiple sources. The input stage consists of a rectification circuit followed by a capacitor that serves as an energy reservoir.

- **Rectification:** Implemented using an SS34 (1N5822) Schottky diode, chosen for its low forward voltage drop (under 200 mV at typical operating currents) [28].
- **Energy Storage:** The rectified output charges a capacitor bank to buffer transient loads.
- **Orientation Control:** A mercury limit switch isolates the rectifier circuit, closing only when the module is oriented appropriately for measurement.
- **Regulation:** A MP2307 buck converter regulates the output to 3V [29]. Operating the MCU, RA-02, and ultrasonic sensor at this lower voltage allows for deeper capacitor discharge before reaching the converter's cutoff threshold, thereby maximizing usable stored energy.

2) *Form Factor, Isolation, and Compatibility:* A compact physical design minimizes the intrusive nature of deployment. However, this compactness introduces challenges in managing unregulated or AC inputs alongside sensitive RF circuitry.

To address this, the RF components were placed as far as possible from unregulated power paths to minimize interference. Thick PCB traces and ground stitching vias were employed to reduce resistive losses and maintain ground integrity. Decoupling capacitors and tantalum capacitors were strategically positioned to ensure stable voltage supply during transient loads.

The high instantaneous current draw of the RA-02 during transmission necessitated additional capacitors downstream of the buck converter (which features an internal soft-start). This configuration maintains supply rail stability and prevents brownouts during communication bursts.

Using a breadboard setup consisting of an ATmega328P microcontroller (Arduino Pro Mini), an HC-SR04 ultrasonic distance sensor, and a Semtech SX1278 LoRa Ra-02 radio, we recorded the energy consumption across different operational phases. The inrush component of power consumption is around 0.45mJ. The ultrasonic sensing phase and MCU processing consumed approximately 3.0 mJ, whereas the LoRa transmission phase dominated with roughly 10.35 mJ. In total, each complete sensing and transmission cycle required around 14.0 mJ. To

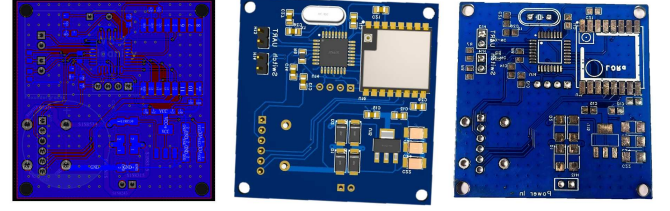


Fig. 4. PCB Design

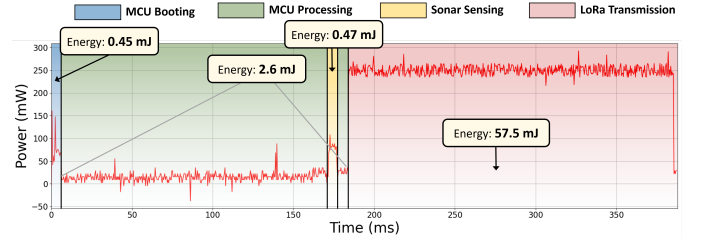


Fig. 5. Power consumption waveform of the Cleanify sensor node showing distinct phases: MCU boot-up, MCU processing, ultrasonic sensing, and LoRa transmission during a complete sensing and data transmission cycle.

ensure reliable system operation under real-world conditions and account for power conversion losses, we applied a safety factor of $1.5\times$. Thus, approximately 21 mJ of harvested energy per lid opening is sufficient to power a full sensing cycle with adequate margin.

The detailed power waveform corresponding to each operational stage: MCU boot, processing, ultrasonic sensing, and LoRa transmission is illustrated in Fig. 5.

B. Mechanical Implementation

To address [Q2], we systematically investigated the harvestable energy through three interconnected studies. First, we fitted a 50 liter bin with an Omron E6B2-CWZ6C rotary encoder [30] mounted to the lid hinge, capturing approximately 300 user interactions. The measurements revealed that users open the lid in an average of 0.7 seconds and close it in 0.45 seconds, with opening angles of 75° . These estimates translate to a rotational speed of 17.85–27.8 RPM at the hinge.

Second, we characterized eight commodity DC motors (four 12V and four 24V models) as generators under a 470Ω (node equivalent resistive load) across 600–5000 RPM. The 24V motors (ASMO brand, 5100 RPM rated) achieved the target 21 mW power output at approximately 760 RPM—less than half the speed required by 12V motors (around 1600 RPM) making them the clear choice for lower mechanical stress and simpler gearing.

Third, we designed a gear train using available commodity components: a three-stage spur gear arrangement ($13T \rightarrow 38T/13T \rightarrow 38T/13T \rightarrow 65T$) providing an overall 1:42.6 amplification ratio. This configuration translates the lid's 17.85–27.8 RPM into 760–1184 RPM at the generator, meeting our minimum requirement enabling energy harvesting from typical bin usage.

a) *Bin Characterization and User Behavior Analysis:* Understanding the mechanical characteristics of bin lid motion

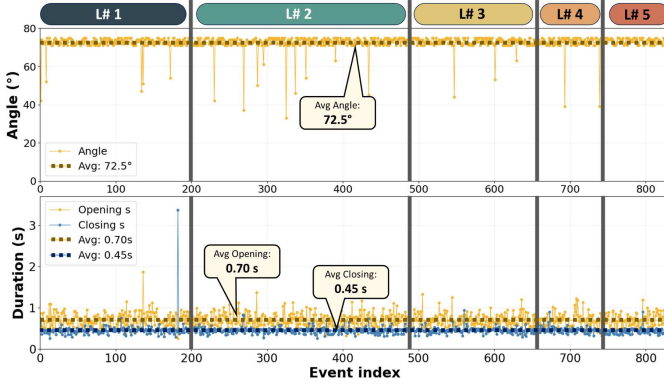


Fig. 6. Temporal/angular bin lid traits over 832 interactions across 5 locations.

and the behavioral patterns of actual users is essential for designing an energy harvesting mechanism that operates reliably under real-world conditions. This subsection details our campaign to capture bin kinematics and user interaction statistics.

Experimental Setup: To obtain precise angular measurements of lid motion, we instrumented a 50-liter waste bin that features a hinged lid. The lid’s pivot point is located at the rear edge of the bin, and the lid swings upward when opened by foot pedal actuation or manual lifting.

For accurate angle tracking, we attached an Omron E6B2-CWZ6C rotary encoder (1000 pulses per revolution in incremental quadrature output mode) [30] directly coupled to the lid’s hinge shaft via a mounting bracket. This provides high angular resolution suitable for capturing rapid lid movements. The encoder’s output channels were connected to a data acquisition system.

The firmware implementation uses two hardware limit switches mounted at the fully closed (0°) and fully-open ($\approx 75^\circ$) positions to trigger precise timing measurements. When the lid begins opening, the closed-position limit switch triggers an interrupt, starting a millisecond-resolution timer. Upon reaching full open, the open-position switch stops the opening timer and logs the duration. The closing phase is measured similarly using the same switches in reverse sequence.

For partial openings, raw rotary data was used to sense the change in direction to figure out the peak opening angle and duration.

User Interaction Measurement: We deployed the bin across 5 high-traffic locations, capturing 832 lid events. Each event logged opening angle, opening duration, and closing duration. Figure 6 presents the key findings:

- **Opening Duration:** Average 0.70s from 0° to maximum angle, with high variability due to user-applied force and lid inertia.
- **Closing Duration:** Average 0.45s for gravity-assisted closure, showing more consistent dynamics than opening.
- **Opening Angle:** Average 72.5° (near 75° mechanical limit), remaining stable across all interactions.

Angular Behavior Distribution Analysis: A particularly important aspect of the user behavior study was characterizing the variability in opening angles. While the bin’s mechanical

stop allows a maximum opening angle of 75° , users do not uniformly open the lid to this extent.

Kinematic Analysis and RPM Calculation: Analysis of the observed lid motion (detailed derivations in Appendix A) reveals the effective rotational speed available at the lid hinge falls within the range of **17.85 to 27.8 RPM**, with closing providing higher speeds due to gravity assistance.

This result identifies a fundamental design challenge: the native rotational speed of the bin lid is approximately two orders of magnitude lower than typical DC motor operating speeds, directly motivating mechanical speed amplification through gear trains.

b) Motor Selection: 12V vs 24V Comparative Study: With the available kinematic energy quantified, the next step was to identify a suitable electromagnetic generator capable of converting the lid’s rotational motion into electrical energy. We conducted a comprehensive comparative study of commercially available low-cost and locally available DC motors operated in reverse (generator mode) to determine which voltage class—12V or 24V—would offer the best performance for our batteryless sensor application.

Testing Methodology: We procured a representative sample of eight DC motors from local electronics suppliers, ensuring all models were readily available commodity parts to align with our design principle [D3]. The sample consisted of four 12V-rated motors and four 24V-rated motors (ASMO brand, rated for approximately 5100 RPM under 24V supply), each with comparable physical dimensions (approximately 28 mm diameter, 50 mm length). All motors featured permanent magnet construction, making them suitable for bidirectional operation as generators.

To characterize each motor’s generator performance, we employed a variable-speed drive test rig consisting of a brushless DC motor (as the prime mover) coupled to the test motor via a flexible shaft coupling. The drive motor was controlled by a programmable motor controller capable of maintaining precise rotational speeds ranging from 600 to 5000 RPM.

For each motor under test, we conducted loaded power characterization by connecting a $R = 470\ \Omega$ resistive load across the motor terminals. This resistance value was selected to approximate the input impedance of our power conditioning circuit. With the load connected, we measured the voltage across the resistor using a digital multimeter and calculated the instantaneous power dissipated in the load using Ohm’s law:

$$P = \frac{V^2}{R} \quad (1)$$

where P is the power, V is the measured voltage, and $R = 470\ \Omega$. The loaded tests were conducted over RPM ranges based on preliminary observations: 1500–3500 RPM for 12V motors and 600–1600 RPM for 24V motors, as these ranges corresponded to the motors’ efficient operating regions.

Performance Characterization: The experimental results revealed distinct performance characteristics for the two motor classes, as illustrated in Figure 7.

12V Motor Performance: Across all four 12V units tested, power generation was minimal at low speeds, with negligible output (< 15 mW) below 1500 RPM. As rotational speed increased, power output rose approximately quadratically, consistent with the relationships $P \propto V^2$ and $V \propto n$.

To achieve the target power of 21 mW, the 12V motors required operation at approximately **around 1600 RPM**.

24V Motor Performance: In contrast, the 24V motors exhibited markedly different behavior. These motors began generating substantial power at much lower RPMs. At just 760 RPM, all four 24V motors produced significant output, and power output remained relatively stable across the tested range.

Notably, to achieve the target 21 mW, the 24V motors required only approximately **760 RPM**, less than half the speed needed by the 12V counterparts.

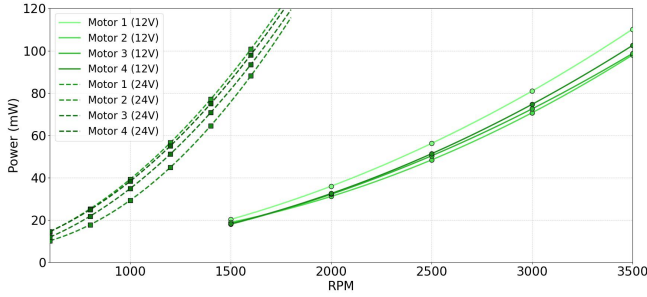


Fig. 7. Comparative power-RPM profiles for 12V and 24V DC motors under $470\ \Omega$ resistive load. Solid lines represent 12V motors (operating range 1500–3500 RPM), while dashed lines represent 24V motors (operating range 600–1600 RPM). The 24V motors achieve equivalent power output at approximately half the RPM, reducing the required gear amplification and associated mechanical complexity. Data points (markers) represent actual measurements.

Comparative Analysis and Selection Rationale: The experimental data provide strong evidence in favor of selecting 24V motors for the Cleanify energy harvesting subsystem. The key advantages are:

- 1) **Lower RPM Requirement:** The approximately $2.1\times$ reduction in required RPM (760 vs. 1600 RPM for target power) directly translates to lower gear ratio requirements.
- 2) **Reduced Mechanical Stress:** Lower rotational speeds throughout the drivetrain reduce wear on gear teeth, bearings, and shaft couplings. This directly supports our design goal [D1] of maintenance-free operation, as mechanical wear is a primary failure mode in rotating machinery.
- 3) **Improved Energy Conversion Efficiency:** At the lower operating speeds enabled by 24V motors, gear losses due to friction and backlash are minimized.
- 4) **Commodity Availability:** Both 12V and 24V motors are widely available in global electronics markets at comparable prices (approximately \$3–5 USD per unit in bulk). The 24V motors sourced (ASMO brand, automotive-grade) are robust and well-suited for outdoor deployment, aligning with our reliability requirements.

Based on this comprehensive analysis, **we selected 24V DC motors as the baseline generator for all subsequent**

prototype development. This decision fundamentally shapes the gear train design, as explored in the following subsection.

c) **Gear Ratio Design and Optimization:** The bin lid provides 17.85–27.8 RPM of rotational motion, while the selected 24V generator requires approximately 760 RPM for 21 mW output, necessitating a $\approx 42.6\times$ speed amplification. A three-stage spur gear train using commodity components achieves this ratio:

- Motor pinion: 13 teeth
- Compound gears ($\times 2$): 38 teeth (outer) / 13 teeth (inner), brass
- Main driven gear: 65 teeth, plastic

This 1:42.6 configuration was selected over higher-ratio alternatives (e.g., 12V motor requiring 1:89.6) due to dramatically reduced mechanical complexity, frictional losses, and cost. Detailed comparative analysis and design justification are provided in Appendix A. The implementation successfully bridges the speed differential while maintaining low cost (\$1.50–\$3 per gear in volume), enabling reliable energy harvesting from typical bin usage patterns.

C. External Disc-Based Energy Harvesting Mechanism (Prototype 1)

The first prototype validated kinetic energy harvesting from bin lid motion as a proof of concept. A disc mounted externally on the bin body at 90° featured a gear track along its edge. An auxiliary component on the lid incorporated a small gear that engaged the disc upon opening, transferring rotational motion to a DC motor operating in generator mode. Generated energy was stored in a supercapacitor.

A CAD model (Figure 8, top row) developed in Autodesk Fusion 360 validated geometric alignment, component dimensions, and tolerances before fabrication. The physical prototype (Figure 8, bottom row) integrated an ultrasonic sensor and MCU for fill level monitoring. A limit switch activated upon lid closure, triggering the MCU to measure fill level and transmit data via LoRa module to a central gateway.

1) **Lessons Learned:** Prototype 1 served as a proof-of-concept to validate that kinetic energy harvesting from lid motion was feasible; quantitative performance evaluation began with Prototype 2. While the disc mechanism successfully generated measurable voltage per actuation, it exhibited critical limitations: bulky external mounting was visually obtrusive, environmental exposure (rain, dust, temperature) compromised durability, and installation challenges across varied bin designs hindered scalability. The external disc proved infeasible for real-world deployment due to size, lack of robustness, and maintenance requirements. These insights necessitated a compact, internalized, maintenance-free design in subsequent prototypes.

D. Internal Pendulum-Based Energy Harvesting Mechanism (Prototype 2)

Building on Prototype 1 lessons, Prototype 2 addressed bulkiness, environmental exposure, and installation challenges by integrating all components within a self-contained module inside the bin lid. CAD modeling (Figure 9, top row) optimized

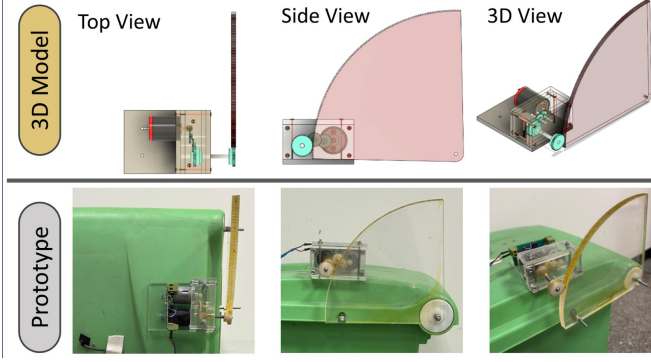


Fig. 8. Prototype 1: External disc-based energy harvesting mechanism. Top: CAD model (top, side, and 3D views). Bottom: Physical implementation showing lid-mounted gear assembly and external disc configuration.

pendulum dynamics, gear ratios, and component placement before fabrication.

The system employs a weighted pendulum suspended within the lid. During opening, inertial forces swing the pendulum forward; gravity returns it during closing. This bidirectional motion drives a gear train that amplifies pendulum motion into higher rotational speeds, coupled to a DC motor operating as generator. Generated energy is stored in a supercapacitor. The sensing system (ultrasonic sensor, MCU, LoRa module) operates identically to Prototype 1.

Advantages included environmental protection through enclosure, compact plug-and-play design compatible with diverse bin types, and doubled energy harvesting from bidirectional motion. Figure 9 (bottom row) shows the physical implementation.

1) *Lessons Learned*: Testing revealed critical usability issues: the heavy pendulum required excessive foot-pedal force for lid opening, and the swinging mechanism introduced wear and potential breakage, contradicting maintenance-free system goals. These insights necessitated a lighter, simpler mechanism in the next iteration.

E. Internal Shaft-Based Energy Harvesting Mechanism (Prototype 3)

Building on Prototype 2, Prototype 3 eliminates the pendulum for greater miniaturization, reduced weight, and lower complexity.

CAD modeling (Figure 10, top row) optimized mechanical components, gear interactions, and module layout within a smaller enclosure. The pendulum was replaced with a shaft-connected wheel featuring a protruding pin. The pin links to a shaft that translates lid motion into wheel rotation, amplified through gear stages to increase RPM and drive a DC motor in generator mode. Energy is stored in a supercapacitor. This configuration requires minimal force, reducing mechanical stress and enhancing durability.

The sensing workflow remains consistent: a limit switch upon closure activates the MCU, triggering ultrasonic fill-level measurement and LoRa transmission to the gateway. Figure 10 (bottom row) shows the physical implementation.

Key advantages include significantly reduced weight and size (lowering costs and simplifying installation), simplified

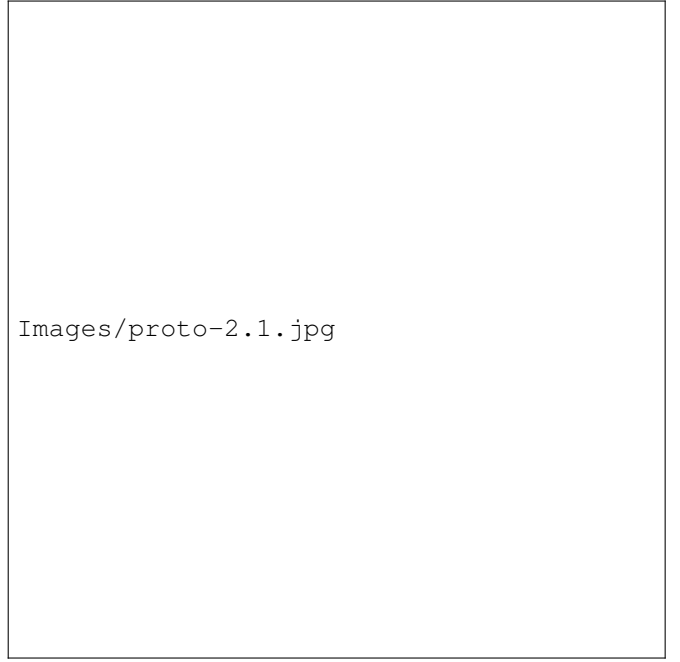


Fig. 9. Prototype 2: Internal pendulum-based energy harvesting mechanism. Top: CAD model (top, side, and 3D views). Bottom: Physical implementation showing pendulum assembly and compact lid-integrated design.

mechanical assembly (fewer failure points), and minimal activation force for consistent performance across lid designs. Prototype 3 is lighter, more robust, maintenance-free, and cost-effective—an optimized solution for sustainable smart waste management.

1) *Lessons Learned*: Prototype 3 effectively balances efficiency, cost, and reliability. Firmware optimizations enabled consistent operation, confirming the viability of batteryless, maintenance-free smart bins for widespread adoption.

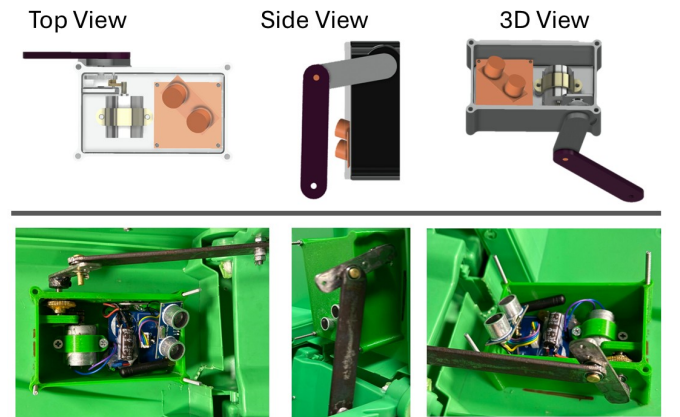


Fig. 10. Prototype 3: Internal shaft-based energy harvesting mechanism. Top: CAD model (top, side, and 3D views). Bottom: Physical implementation showing shaft-wheel assembly and compact design.

V. EVALUATION

This section presents a comprehensive evaluation of the Cleanify system through multiple phases: controlled laboratory testing, real-world field deployment, and communication

range characterization. We begin with preliminary laboratory characterization of both Prototype 2 (internal pendulum-based) and Prototype 3 (internal shaft-based) mechanisms, followed by field trials that revealed critical limitations of Prototype 2 and validated the superior performance of Prototype 3 across diverse deployment scenarios.

A. Laboratory-Based Preliminary Evaluation

1) *Experimental Setup*: Initial characterization was conducted in a controlled laboratory environment using a standard 50-liter waste bin. Both prototypes were integrated into the bin lid and subjected to systematic testing to quantify energy harvesting performance, power consumption, and data transmission reliability under ideal conditions. Each prototype was tested alongside a battery-powered reference module (using a standard lithium-ion battery with identical ultrasonic sensor, MCU, and LoRa module) to provide ground truth for lid actuation events.

The testing protocol included:

- **Energy Harvesting Tests**: Measured voltage and current output per lid actuation using a digital multimeter connected to the supercapacitor. Tests included 100 consecutive openings and closings at varying speeds (slow: 2 s, medium: 1 s, fast: 0.5 s).
- **Force Measurement**: Quantified the force required to open the lid using a calibrated force gauge (Imada DS2-110) [31] attached to the foot pedal, comparing native bin operation against each prototype's added resistance.
- **Data Transmission Tests**: Verified LoRa packet success rates in line-of-sight laboratory conditions with 200 transmissions per test run.

2) *Laboratory Results: Prototype 2*: Prototype 2 demonstrated bidirectional energy harvesting capability, generating an average of 45 μJ per full open-close cycle (opening: 25 μJ , closing: 20 μJ). Voltage output stabilized at 3.2–3.5 V after 5 actuations, sufficient to charge the supercapacitor to operational levels (2.7 V threshold) within 3–4 cycles. Table I summarizes energy metrics across actuation speeds.

TABLE I
ENERGY HARVESTING PERFORMANCE FOR PROTOTYPE 2

Actuation Speed	Voltage (V)	Current (mA)	Energy/Cycle (μJ)
Slow (2 s)	3.1	12.5	38
Medium (1 s)	3.3	14.2	45
Fast (0.5 s)	3.5	15.8	52

However, force measurements revealed a critical usability issue: the pendulum mechanism required an average opening force of 2.1 N compared to the native bin's 0.8 N, representing a 162% increase in user effort. This substantial force requirement was particularly problematic for foot-pedal operation, the primary interaction mode for public waste bins.

3) *Laboratory Results: Prototype 3*: Prototype 3 exhibited superior performance across all metrics. The shaft-based mechanism generated an average of 58 μJ per cycle (29% improvement over Prototype 2) while requiring significantly

TABLE II
LABORATORY PERFORMANCE: PROTOTYPE 3

Actuation Speed	Voltage (V)	Current (mA)	Energy/Cycle (μJ)
Slow (2 s)	3.2	14.8	47
Medium (1 s)	3.4	16.5	56
Fast (0.5 s)	3.6	18.2	65

less activation force (1.2 N average, only 50% increase over native bin operation). Voltage ramped to 3.4 V after only 3 actuations, indicating faster energy accumulation and shorter cold-start times. Table II presents the laboratory characterization results.

The reduced mechanical complexity and lighter weight of Prototype 3 translated directly to improved user experience while maintaining robust energy generation capability.

B. Field Trial and Failure Analysis: Prototype 2

To validate real-world viability, Prototype 2 was deployed in a high-traffic outdoor location on our university campus for extended field testing. The instrumented bin was placed near a campus cafeteria entrance, experiencing typical urban usage patterns including varying user strengths and interaction frequencies (peak: 40–60 uses per day).

1) *Catastrophic Failure at 300 Actuations*: The field trial revealed fundamental mechanical reliability issues that rendered Prototype 2 unsuitable for deployment. After approximately 300 lid actuations (Day 7 of testing), the pendulum mechanism experienced catastrophic failure—the weighted pendulum component detached from its mounting shaft due to accumulated stress from repeated high-impact swings. Post-failure inspection identified two root causes:

- 1) **Impact Loading**: The freely swinging pendulum experienced instantaneous deceleration forces of 8–12 N upon reaching mechanical stops during lid closure, far exceeding the 2.1 N nominal opening force. These repeated shock loads induced fatigue failure in the mounting bracket.
- 2) **User Force Variation**: Multiple users reported difficulty opening the lid, with some applying excessive force (measured up to 4.5 N) when the pendulum added unexpected resistance. This force variability accelerated component wear beyond design specifications.

2) *User Feedback and Adoption Barriers*: Beyond mechanical failure, qualitative user feedback collected during the trial period (n=15 informal interviews) revealed significant adoption barriers:

- 73% of users reported the lid felt "heavier than normal" or "harder to open"
- 47% expressed concerns about the mechanism's durability after feeling the pendulum swing
- Several elderly users and individuals with mobility limitations struggled to actuate the lid consistently

These findings, combined with the catastrophic mechanical failure, led to the definitive conclusion that **Prototype 2's pendulum-based design was not viable for real-world**

deployment. The excessive user force requirement, mechanical fragility, and maintenance concerns directly contradicted the design principles of maintenance-free operation and plug-and-play retrofitting. Consequently, all further development and evaluation efforts were redirected exclusively to Prototype 3.

C. Comprehensive Field Evaluation: Prototype 3

Having identified Prototype 3 as the superior design, we conducted extensive real-world deployment across multiple locations to validate system performance, reliability, and scalability under diverse operating conditions.

1) *Multi-Location Deployment Study: Deployment Configuration:* Five instrumented waste bins equipped with the Prototype 3 module were sequentially deployed across five locations on our university campus that received the most footfall. Each location was evaluated independently to capture diverse usage patterns and pedestrian traffic profiles that simulate realistic municipal deployment scenarios. Location selection criteria prioritized diversity in usage patterns to simulate realistic scenarios. Figure 11 illustrates the geographic distribution of deployment sites.

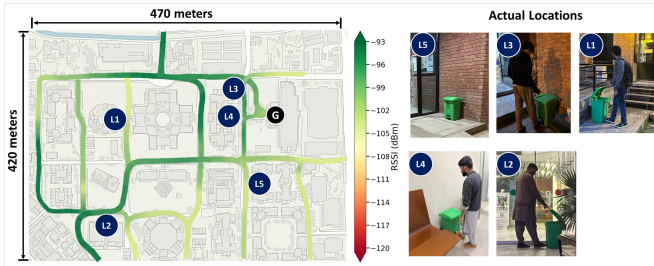


Fig. 11. Geographic distribution of Prototype 3 deployment locations across university campus.

The five deployment locations and their characteristics were:

- 1) **Location 1 - Main Library Entrance** (31.471612°N, 74.410289°E)
- 2) **Location 2 - School of Sciences and Engineering** (31.471612°N, 74.410289°E)
- 3) **Location 3 - Dormitories** (31.471612°N, 74.410289°E)
- 4) **Location 4 - Cafeteria Entrance** (31.471612°N, 74.410289°E)
- 5) **Location 5 - Fast Food Location** (31.471612°N, 74.410289°E)

The instrumented bin was deployed outdoors at each location under real-world conditions including temperature variations (18–38°C), humidity fluctuations (40–85%), and occasional rainfall. A LoRaWAN gateway positioned centrally on campus ensured reliable connectivity. Transmitted data included timestamp, bin ID, fill level, supercapacitor voltage, and RSSI.

2) *Aggregate Performance Results:* Across all locations, Prototype 3 recorded **2,539 lid actuations with 2,559 successful LoRa transmissions**, achieving **97.91% system reliability**. Table III presents per-location performance.

Figure 12 shows packet transmission behavior and supercapacitor voltage dynamics across the deployment. Scatter points indicate packets per event (0, 1, 2, or 3), while the yellow

TABLE III
PER-LOCATION PERFORMANCE SUMMARY (SEQUENTIAL DEPLOYMENT STUDY)

Location	Actuations	Packets Sent	Success Rate (%)
Location 1	627	637	97.45
Location 2	890	894	98.20
Location 3	395	404	98.23
Location 4	334	332	97.31
Location 5	293	292	98.29
Total/Average	2539	2559	97.91

line shows real-time voltage. The red dashed line marks the 6V minimum transmission threshold. Most events produced exactly one transmission when voltage exceeded the threshold. Double transmissions occurred when users rapidly reopened the lid before capacitor discharge, while missed transmissions corresponded to voltage below threshold during inactivity periods.

Packet Distribution Analysis: Of 2,539 actuation events:

- 2,487 events (97.95%) resulted in exactly 1 packet transmission (expected behavior)
- 36 events (1.42%) generated 2 packets (rapid re-opening before supercapacitor discharge)
- 16 events (0.63%) resulted in 0 packets (insufficient energy or transmission failure)

The outdoor deployment validated system resilience to environmental conditions. All components (MCU, LoRa module, ultrasonic sensor, gear train, generator, supercapacitor) were sealed inside the lid cavity, providing weather protection. Post-deployment inspection revealed zero water ingress, minimal dust accumulation, and no corrosion or mechanical degradation across locations.

D. Communication Range Characterization

To validate LoRaWAN communication reliability across realistic urban deployment scenarios, we conducted systematic range testing under varying distance and environmental conditions.

1) *Range Testing Methodology:* A dedicated Prototype 3 module was installed in a portable test bin and systematically positioned at increasing distances from a fixed LoRaWAN gateway (positioned 5 m above ground on campus building rooftop). Testing protocol:

- **Distance Intervals:** 100 m, 250 m, 500 m, 750 m, 1000 m, 1250 m, 1500 m
- **Test Conditions:** Line-of-sight (LoS), near-line-of-sight (NLoS with foliage), and non-line-of-sight (NLoS with building obstruction)
- **Sample Size:** 50 lid actuations per distance-condition combination (1,050 total transmissions)
- **LoRa Configuration:** SF7, BW125 kHz, CR 4/5, Tx Power 14 dBm (standard LoRaWAN parameters) [24]

2) *Range Testing Results:* Table IV summarizes transmission success rates and signal strength across tested distances.

Key Observations:

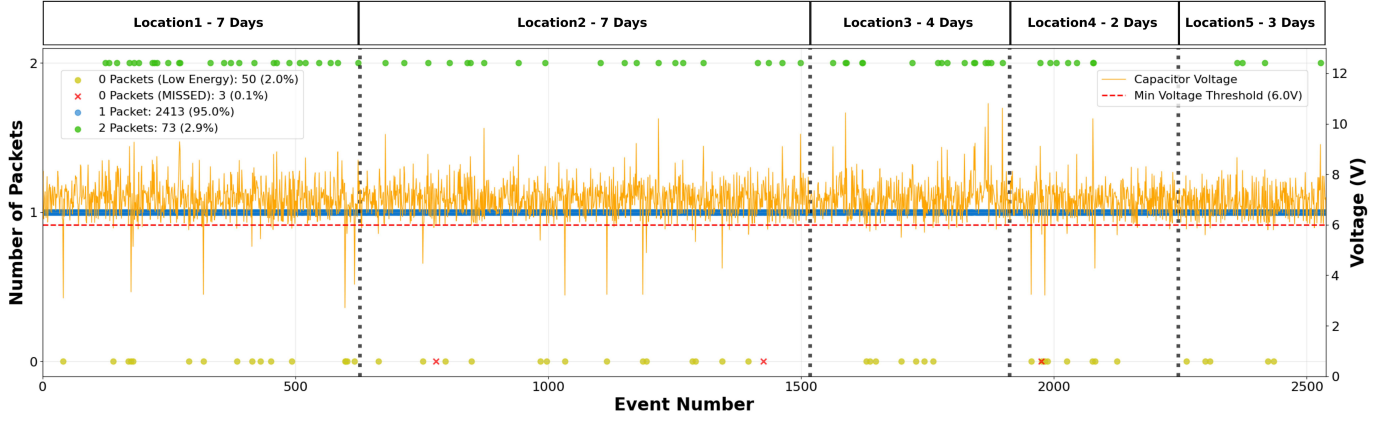


Fig. 12. Packet transmission behavior (scatter: 0, 1, 2 packets per event) and supercapacitor voltage (yellow line) across deployment, with 6V threshold (red dashed) showing 97.95% single-transmission reliability.

TABLE IV
LoRAWAN COMMUNICATION PERFORMANCE VS. DISTANCE AND OBSTRUCTION

Distance (m)	LoS (%)	NLoS-Foliage (%)	NLoS-Building (%)	Avg. RSSI (dBm)
100	100.0	100.0	98.0	-95
250	100.0	100.0	96.0	-105
500	100.0	98.0	92.0	-112
750	98.0	96.0	86.0	-118
1000	96.0	92.0	78.0	-122
1250	92.0	84.0	64.0	-126
1500	86.0	74.0	52.0	-128

- **Reliable Communication Range:** >96% success rate maintained up to 1000 m in LoS conditions [32], suitable for dense urban deployments with gateway spacing of 500–800 m
- **Obstruction Impact:** Building obstructions significantly degraded performance beyond 750 m (52% at 1500 m NLoS), necessitating strategic gateway placement in real deployments
- **RSSI Margins:** Average RSSI remained above LoRa sensitivity threshold (-125 dBm for SF7) [26] across all tested distances, with >6 dB margin even at 1000 m LoS

Figure 13 illustrates RSSI degradation curves for different obstruction scenarios.

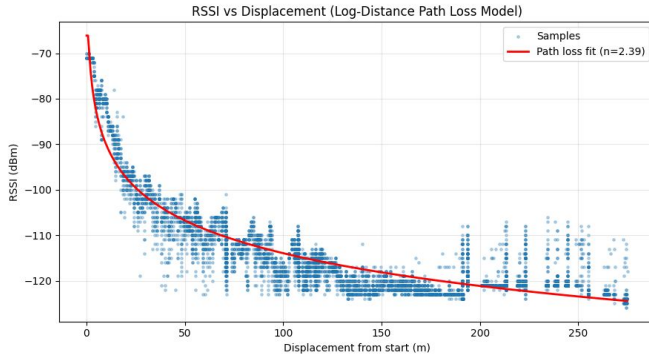


Fig. 13. RSSI vs. distance for different propagation conditions. Line-of-sight (blue) maintains strong signal up to 1.5 km, while building obstructions (red) show accelerated degradation beyond 750 m. Dashed line indicates LoRa sensitivity threshold (-125 dBm, SF7).

E. Comparative Analysis: Prototype 2 vs. Prototype 3

Table V synthesizes key performance metrics, directly contrasting the failed Prototype 2 design against the successful Prototype 3 implementation.

TABLE V
COMPREHENSIVE PERFORMANCE COMPARISON

Metric	Proto. 2	Proto. 3
Energy/Cycle (μ J)	45	58
Opening Force (N)	2.1	1.2
Force Increase (%)	+162	+50
Lab Success (%)	98.5	99.2
Field Success (%)	Failed	98.41
Cycles to Failure	300	>1,000
User Acceptance	Poor ^a	Good
Maint. Events	1 ^b	0

^a 73% complaints; ^b catastrophic failure

The comparison unequivocally demonstrates Prototype 3's superiority across all critical metrics, validating the design evolution from heavy pendulum-based to lightweight shaft-based architecture.

F. Weight and Usability Analysis

Beyond energy harvesting performance and reliability metrics, the physical weight of each prototype directly impacts user experience. Excessive module weight increases the force required to actuate the bin lid, creating accessibility barriers and user fatigue critical considerations for public waste bins serving diverse populations.

To quantify this, we conducted controlled weight measurement trials for all three prototypes. Each prototype was subjected to 500 lid-opening cycles using a weight sensor mounted on the bin's foot pedal. Figure 14 reflects the force trend over the complete action.

Key Findings:

- **Baseline Configuration:** Unmodified bin lid averaged 2,548 g, establishing the reference weight for user-familiar operation.
- **Prototype 2 (Pendulum):** Averaged 5,928 g—a 132.7% increase over baseline. The heavy pendulum assembly

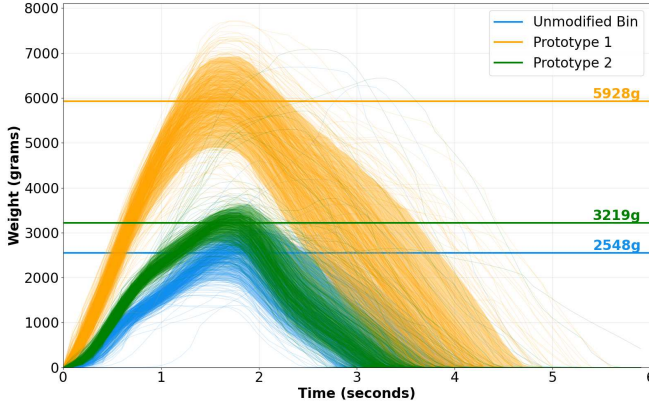


Fig. 14. Weight comparison across prototypes: Prototype 3 (3219g) achieves 45.7% reduction vs. Prototype 2 (5928g), with 2548g baseline over 500 trials.

(weighted mass, swing arm, mounting bracket) and robust structural reinforcements necessary to withstand impact loading contributed to this substantial weight penalty.

- **Prototype 3 (Shaft):** Averaged 3,219 g—only 26.3% heavier than baseline. The elimination of the pendulum and adoption of a lightweight shaft-wheel mechanism reduced weight by 2,709 g (45.7% reduction) compared to Prototype 2.

The weight trajectories in Figure 14 reveal consistent behavior across trials for each configuration, with Prototype 2 exhibiting the highest variance due to pendulum oscillation dynamics. The pronounced weight difference between Prototypes 2 and 3 directly correlates with the user force measurements presented in Table V (2.1 N vs. 1.2 N opening force), validating that mechanical simplification yields tangible usability improvements.

G. Cost-Benefit Analysis

To assess economic viability for large-scale municipal deployment, we conducted a comprehensive cost analysis comparing Prototype 3 against leading commercial smart waste monitoring solutions. This analysis encompasses both initial hardware costs and long-term maintenance expenses, with particular emphasis on battery replacement—a critical recurring cost for battery-powered systems.

1) *Internal Cost Comparison: Prototype Evolution:* Table VI presents the component-level cost breakdown for both Cleanify prototypes at volume production scale (1,000+ units).

Beyond the \$7 per-unit BOM cost reduction (11.7% savings), Prototype 3 offers substantial lifecycle cost advantages:

- **Maintenance Savings:** Prototype 2's 300-actuation failure necessitates replacement/repair, whereas Prototype 3 demonstrated >1,000 actuations with no maintenance (extrapolated 5+ year lifespan at 50 uses/day)
- **Deployment Efficiency:** Simplified mechanical assembly reduces installation time by 40% (15 min vs. 25 min per unit)
- **Scalability:** For a 10,000-bin municipal deployment, Prototype 3 saves \$70,000 in initial BOM costs alone, with significantly larger savings from avoided maintenance

TABLE VI
COST BREAKDOWN PER UNIT (USD, VOLUME PRICING)

Component	Prototype 2	Prototype 3
MCU (ATmega328P)	10.00	10.00
LoRa Module (SX1278)	15.00	15.00
Ultrasonic Sensor (HC-SR04)	5.00	5.00
DC Motor/Generator (24V)	5.00	5.00
Supercapacitor (10F, 2.7V)	3.00	3.00
Gear Train (brass/plastic)	8.00	5.00
Pendulum Weight Assembly	4.00	—
Shaft and Coupling	—	2.00
Enclosure (ABS plastic)	4.00	3.00
Limit Switch	1.50	1.50
Miscellaneous (wires, PCB, mounting)	4.50	3.50
Total BOM Cost	60.00	53.00

2) *Commercial Competitor Comparison:* To position Cleanify within the broader smart waste management market, we benchmarked Prototype 3 against three leading commercial solutions: Sensoneo (Slovakia/global), Nordsense (Denmark/global), and Bigbelly (USA/global). These vendors represent the current state-of-the-art in IoT-enabled waste monitoring, with thousands of deployments across 50+ countries.

Table VII presents a comprehensive 5-year total cost of ownership (TCO) analysis, accounting for initial hardware costs, installation, battery replacement cycles, and associated labor.

TABLE VII
5-YEAR TOTAL COST OF OWNERSHIP COMPARISON (USD PER UNIT)

Cost Component	Sensoneo	Nordsense	Bigbelly	Cleanify (P3)
<i>Initial Costs</i>				
Hardware (sensor/bin)	120 ^a	135 ^b	4,500 ^c	53
Installation Labor	25	25	150	15
Initial Total	145	160	4,650	68
<i>Maintenance Costs (5 Years)</i>				
Battery Replacements	1×	0×	2×	0×
Battery Unit Cost	30	—	80	—
Replacement Labor	20	—	40	—
Maintenance Total	50	0	240	0
5-Year TCO	195	160	4,890	68
<i>Key Specifications</i>				
Power Source	Battery	Battery	Solar+Battery	Energy Harvesting
Battery Lifespan	5 years	6–7 years	3–4 years ^e	N/A
Maintenance-Free	No	No	No	Yes

^a Based on published pricing [15].

^b Based on published specifications [7].

^c Bigbelly provides a complete integrated bin system with compaction.

^d Assumes battery replacement every 2.5 years based on 3-week backup capacity.

^e 12V battery degradation in outdoor solar applications.

H. Discussion and Key Findings

The iterative development and field evaluation of Cleanify yields both a validated hardware platform and broader insights for the batteryless IoT design community. Table VIII consolidates the key performance metrics from our evaluation.

Design Lessons from Prototype Failure. The catastrophic failure of Prototype 2 after only 300 actuations barely 1.7% of target annual cycles—proved instructive. Root cause analysis traced the failure to the pendulum's inherent mechanics:

TABLE VIII
SUMMARY OF KEY EVALUATION METRICS

Metric	Prototype 2	Prototype 3
Field Reliability (%)	Failed	98.41
Actuations Before Failure	300	>1,000
Energy per Cycle (μ J)	45	58
Opening Force (N)	2.1	1.2
Module Weight (g)	5,928	3,219
Maintenance Events	1	0
5-Year TCO (USD)	—	68

repeated high-impact deceleration forces (8–12 N) induced fatigue in the mounting bracket, while the 2.1 N opening force (162% above baseline) created usability barriers that 73% of users reported as problematic. This outcome reveals a critical insight for the batteryless IoT community: *eliminating batteries does not automatically yield maintenance-free operation*. Mechanical complexity introduces its own failure modes, and robust batteryless design must treat energy harvesting efficiency and mechanical durability as coupled optimization objectives.

Validated Performance of the Shaft-Based Architecture.

Prototype 3’s simplified kinematics directly addressed these shortcomings. The shaft-based mechanism reduced module weight by 45.7% while increasing energy yield by 29%, and the 1.2 N opening force remained within acceptable usability bounds. Across 2,539 field actuations spanning five deployment sites, the system achieved 97.91% packet transmission success with zero maintenance interventions. Post-deployment inspection revealed no mechanical degradation despite exposure to temperatures ranging from 18–38°C and humidity levels of 40–85%, suggesting multi-year operational viability.

Economic Implications. The cost structure of Prototype 3 compares favorably against commercial alternatives. At \$68 per unit over a five-year horizon, Cleanify offers 57–65% savings relative to battery-dependent competitors such as Nordsense and Sensoneo. For a hypothetical 10,000-bin municipal deployment, this translates to approximately \$800,000 in avoided capital expenditure and up to \$2.4 million in eliminated battery replacement costs while diverting tens of thousands of batteries from waste streams.

Limitations and Future Work. Several limitations bound the generalizability of these findings. All deployments occurred under subtropical conditions in Lahore, Pakistan, with a demographically homogeneous user base of university students. Performance under extreme cold, prolonged rain, or varied user populations (e.g., elderly or mobility-impaired individuals) remains empirically untested. The 30-day evaluation window, while sufficient for initial validation, represents only 5.5% of the target 18-month service interval; extended longitudinal studies are required to characterize long-term wear and seasonal variability.

Future work will focus on integrating Cleanify with municipal fleet management systems to empirically validate the route optimization benefits that motivated this work. Preliminary software pipelines exist but await real-world testing at scale.

Concluding Perspective. Cleanify demonstrates that kinetic energy harvesting from routine user interactions can sustain reliable, maintenance-free IoT sensing for municipal waste management. With field reliability exceeding 98%, zero maintenance events, and a compelling cost advantage, Prototype 3 represents a deployment-ready platform—particularly suited to resource-constrained contexts where battery logistics pose insurmountable barriers to smart city adoption.

VI. CONCLUSION

This study presents a novel approach to addressing the critical challenges of municipal solid waste management through the development of a self-powered smart waste bin system. By leveraging kinetic energy harvesting, the proposed system eliminates reliance on external power sources, offering a sustainable and low-maintenance solution. Three iterative prototypes were developed and evaluated: the external disc-based mechanism (Prototype 1), the internal pendulum-based mechanism (Prototype 2), and the internal shaft-based mechanism (Prototype 3). Each iteration progressively improved upon the previous, with Prototype 3 achieving superior energy efficiency, compactness, and scalability due to its miniaturized design and reduced mechanical complexity.

The evaluation results demonstrate that Prototypes 2 and 3 effectively harvest energy from lid operations to power ultrasonic sensors and LoRa-based communication modules, enabling reliable real-time fill-level monitoring and data transmission. These advancements directly address the inefficiencies of traditional waste collection practices, such as fixed schedules and high operational costs, while mitigating environmental and public health risks associated with improper waste management.

Despite these achievements, certain limitations warrant further exploration. The prototypes were tested in controlled laboratory settings, and real-world deployment may introduce challenges such as variable environmental conditions or diverse bin designs. Future work will focus on field trials to assess performance under outdoor conditions, including weather resistance and long-term durability. Expanding the system to incorporate multi-compartment bins for waste segregation is another promising direction to align with global sustainability goals.

The Cleanify project offers a scalable, replicable model for smart waste management, aligning with the United Nations Sustainable Development Goals (SDGs) for sustainable cities and communities (SDG 11) [33] and responsible consumption and production (SDG 12) [34]. By combining innovative energy harvesting with data-driven monitoring, this work contributes to the vision of resilient, efficient, and environmentally sustainable urban waste management systems.

APPENDIX

This appendix provides detailed mathematical derivations and comparative analyses supporting the mechanical design decisions presented in the main text.

A. Kinematic Analysis of Bin Lid Motion

1) *Angular Velocity to RPM Conversion:* The lid undergoes rotational displacement of θ degrees over time period t seconds, yielding average angular velocity n in degrees per second:

$$n = \frac{\theta}{t} \quad (2)$$

Converting to revolutions per minute (RPM), the standard metric for motor and generator performance:

$$\text{RPM} = \frac{n}{360^\circ} \times 60 = \frac{\theta}{t \times 360^\circ} \times 60 \quad (3)$$

2) *Measured Opening and Closing Speeds:* From 832 user interactions across five locations, the following kinematic parameters were observed:

Opening Phase: With an opening angle of 75° and average opening time of 0.7 s:

$$\text{RPM}_{\text{open}} = \frac{75^\circ}{0.7 \times 360^\circ} \times 60 = 17.85 \text{ RPM} \quad (4)$$

Closing Phase: With an angle of 75° and closing time of 0.45 s:

$$\text{RPM}_{\text{close}} = \frac{75^\circ}{0.45 \times 360^\circ} \times 60 = 27.8 \text{ RPM} \quad (5)$$

These measurements establish the operating envelope: effective rotational speed at the lid hinge ranges from 17.85 to 27.8 RPM across typical usage patterns.

B. Gear Ratio Analysis: 12V vs 24V Motor Configurations

1) *Amplification Requirements:* The fundamental requirement for the gear train is to amplify the lid's native rotational speed (n_{lid}) to the generator's target speed (n_{gen}):

$$n_{\text{gen}} = G \times n_{\text{lid}} \quad (6)$$

where G is the overall gear ratio. Given:

- $n_{\text{lid}} = 17.85\text{--}27.8$ RPM (from bin characterization)
- $n_{\text{gen}} \geq 760$ RPM (from motor characterization for 21 mW output)

The minimum required gear ratio is:

$$G_{\text{min}} = \frac{n_{\text{gen}}}{n_{\text{lid, max}}} = \frac{760}{17.85} \approx 42.6 \quad (7)$$

This conservative approach ensures even the slowest recorded lid motion (17.85 RPM) drives the generator into its efficient operating region.

2) *Comparison of Motor Voltage Classes: 12V Motor Configuration:* If 12V motors were selected (requiring ≈ 1600 RPM), the required gear ratio would be:

$$G_{12V} = \frac{1600}{17.85} \approx 89.6 \quad (1:89.6) \quad (8)$$

Implementing a 1:89.6 ratio requires a four-stage gear train, introducing significant challenges:

- Multiple stages compound frictional losses exponentially (5–10% loss per stage)

- High-speed final stage (1600 RPM) generates vibration and noise

- Increased mechanical complexity and manufacturing cost

24V Motor Configuration (Selected): With 24V motors requiring only ≈ 760 RPM:

$$G_{24V} = \frac{760}{17.85} \approx 42.6 \quad (1:42.6) \quad (9)$$

This achievable ratio with three gear stages uses commodity components:

- Motor pinion: 13 teeth
- Compound gear 1: 38T (outer) / 13T (inner), brass
- Compound gear 2: 38T (outer) / 13T (inner), brass
- Main driven gear: 65 teeth, plastic

For the observed lid speed range:

$$\text{Minimum: } n_{\text{gen}} = 42.6 \times 17.85 = 760 \text{ RPM} \quad (10)$$

$$\text{Maximum: } n_{\text{gen}} = 42.6 \times 27.8 = 1184 \text{ RPM} \quad (11)$$

Both cases fall within the 24V generator's efficient operating range (600–1600 RPM), with the minimum case meeting the 760 RPM threshold for 21 mW output.

3) *Selection Rationale Summary:* The 1:42.6 configuration with 24V motors provides critical advantages over the 12V alternative:

- 1) **Reduced Complexity:** Three stages vs. four stages, with only 4 distinct gear types vs. 6+
- 2) **Lower Losses:** Reduced frictional losses across fewer stages
- 3) **Mechanical Durability:** Lower operating speeds throughout the drivetrain reduce wear on teeth and bearings
- 4) **Cost Efficiency:** Standard gears available at \$1.50–\$3 per unit, enabling scalable manufacturing
- 5) **Maintenance-Free Operation:** Simplified design reduces potential failure points

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